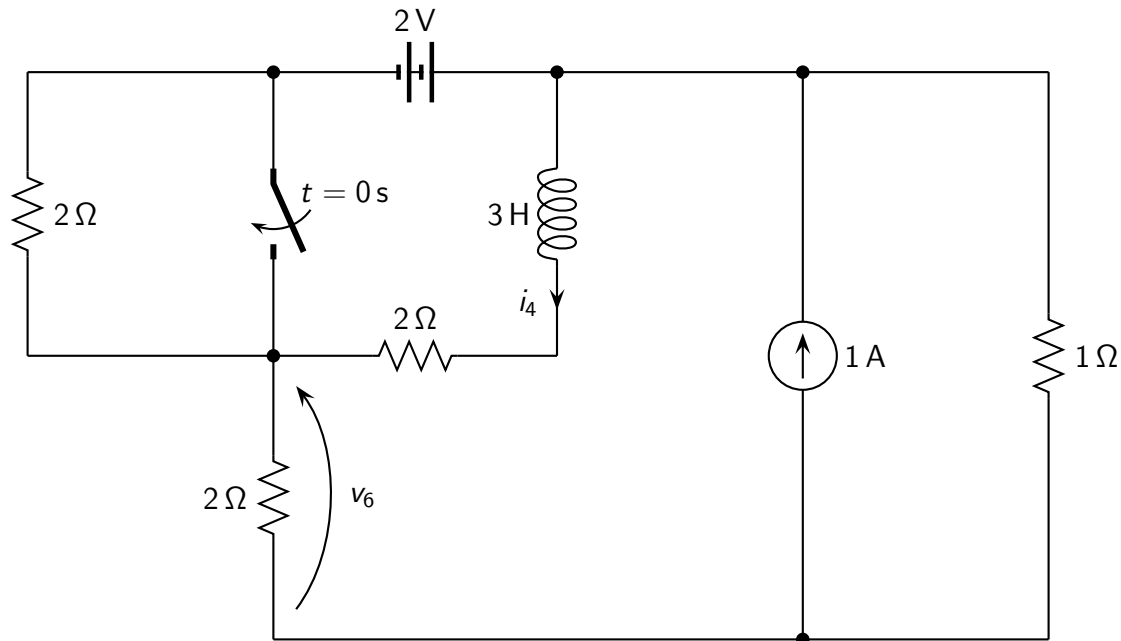


Problem: find the expression of $i_4(t)$, $v_6(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{3}{2} \text{ s}$;
- Initial conditions before switching: $i_4(0^-) = \frac{1}{2} \text{ A}$;
- Initial conditions after switching: $i_4(0^+) = \frac{1}{2} \text{ A}$; $v_6(0^+) = \frac{-2}{3} \text{ V}$;
- Steady-state solution: $i_4(\infty) = 1 \text{ A}$; $v_6(\infty) = \frac{-2}{3} \text{ V}$;
- Solution for $t > 0$:

$$i_4(t) = \left(\frac{-1}{2} e^{-t/\tau} + 1 \right) \text{ A}$$
$$v_6(t) = \frac{-2}{3} \text{ V}$$